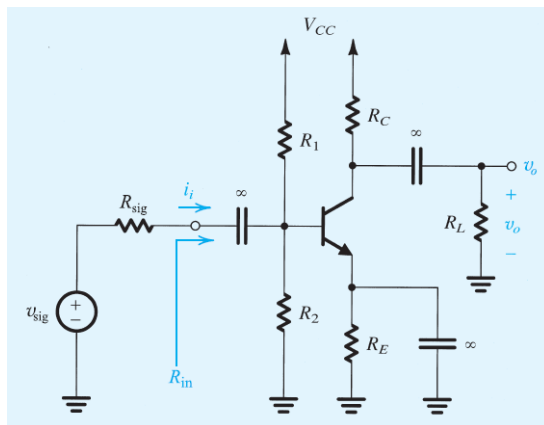


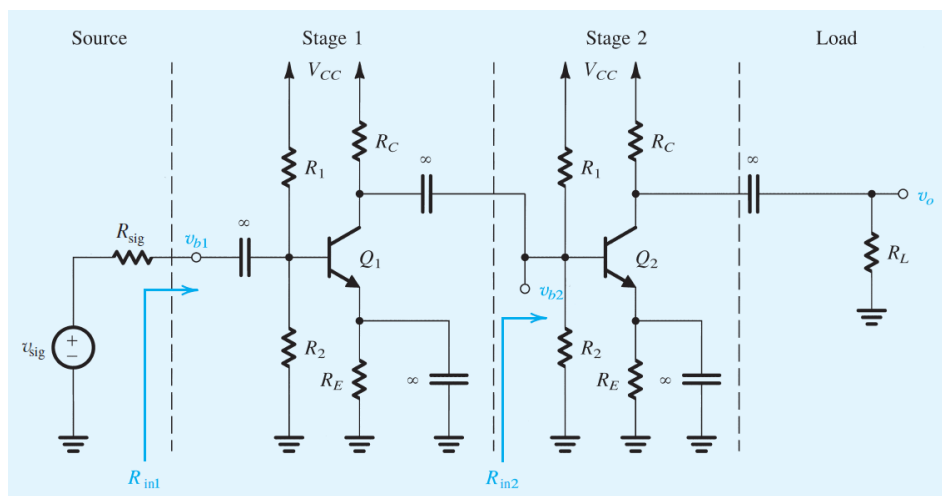
Homework 11

Topic: BJTs Small Signal Analysis

- For the common-emitter amplifier shown below, let $V_{CC}=15\text{ V}$, $R_1=27\text{ k}\Omega$, $R_2=15\text{ k}\Omega$, $R_E=2.4\text{ k}\Omega$, and $R_C=3.9\text{ k}\Omega$. The transistor has $\beta=100$.
 - Calculate the dc bias current I_C .
 - If the amplifier operates between a source for which $R_{sig}=2\text{ k}\Omega$ and a load of $2\text{ k}\Omega$, replace the transistor with its hybrid- π model, and find the values of R_{in} , and the overall voltage gain v_o/v_{sig} .
 - Find R_{out} .
 - Choose some sensible values for the capacitors (think about the role each capacitor plays in this circuit) to estimate the time constants
 - Make a rough sketch of the Bode plot (magnitude only).

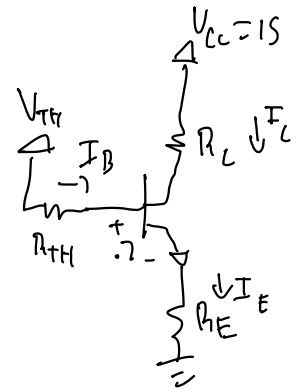
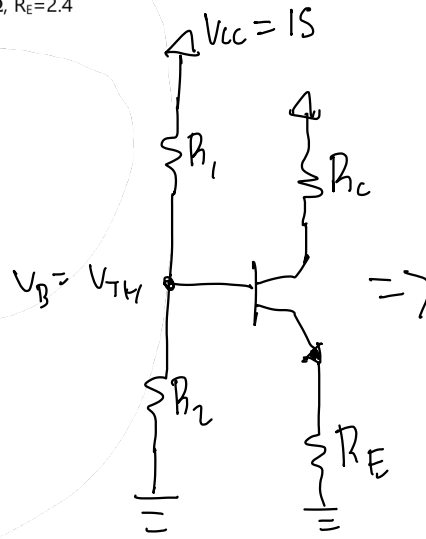
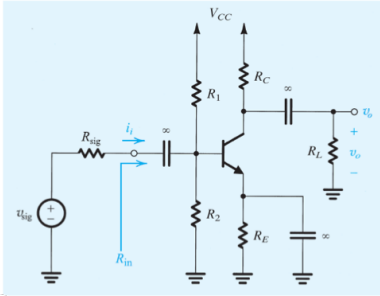


- The amplifier below consists of two identical common-emitter amplifiers connected in cascade. Observe that the input resistance of the second stage, R_{in2} , constitutes the load resistance of the first stage.
 - For $V_{CC}=15\text{ V}$, $R_1=100\text{ k}\Omega$, $R_2=47\text{ k}\Omega$, $R_E=3.9\text{ k}\Omega$, $R_C=6.8\text{ k}\Omega$, and $\beta=100$, determine the dc collector current and dc collector voltage of each transistor.
 - Draw the small-signal equivalent circuit of the entire amplifier and give the values of all its components.
 - Find R_{in1} and v_{b1}/v_{sig} for $R_{sig}=5\text{ k}\Omega$.
 - Find R_{in2} and v_{b2}/v_{b1} .
 - For $R_L=2\text{ k}\Omega$, find v_o/v_{b2} .
 - Find the overall voltage gain v_o/v_{sig} .
 - Find R_{out} .
 - Choose some sensible values for the capacitors (think about the role each capacitor plays in this circuit) to estimate the time constants
 - Make a rough sketch of the Bode plot (magnitude only).



1. For the common-emitter amplifier shown below, let $V_{CC}=15\text{ V}$, $R_1=27\text{ k}\Omega$, $R_2=15\text{ k}\Omega$, $R_E=2.4\text{ k}\Omega$, and $R_C=3.9\text{ k}\Omega$. The transistor has $\beta=100$.

(a) Calculate the dc bias current I_C .



$$V_{TH} = 15 \left(\frac{R_2}{R_1 + R_2} \right) = \frac{75}{14} [\text{V}]$$

$$R_{TH} = R_1 || R_2 = 9.634 \text{ k}\Omega$$

$$V_{TH} - 0 = I_B R_{TH} + 0.7 + I_E R_E$$

$$V_{TH} - 0.7 = I_B (R_{TH} + (\beta + 1) R_E)$$

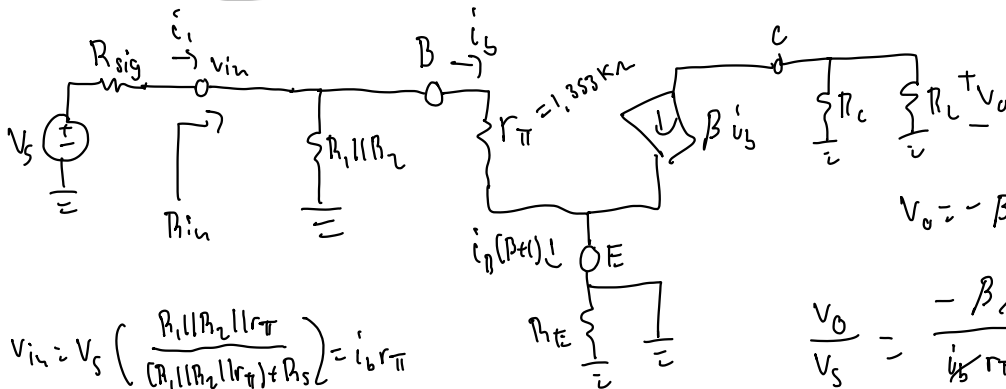
$$\frac{143}{35} = I_B (R_{TH} + (\beta + 1) R_E)$$

$$I_B = \frac{143}{35} \cdot \frac{1}{(R_{TH} + (\beta + 1) R_E)}$$

$$I_B \approx 18.5 \mu\text{A}$$

$$I_C = \beta I_B \approx 1.85 \text{ mA}$$

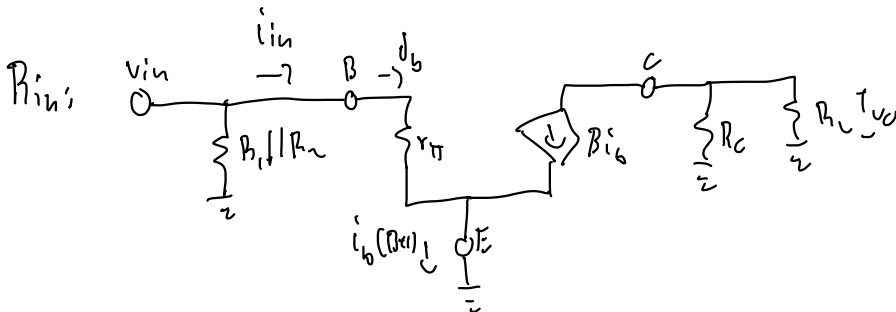
(b) If the amplifier operates between a source for which $R_{sig}=2\text{ k}\Omega$ and a load of $2\text{ k}\Omega$, replace the transistor with its hybrid- π model, and find the values of R_{in} , and the overall voltage gain v_o/v_{sig} .



$$v_o = -\beta i_b (R_C || R_L)$$

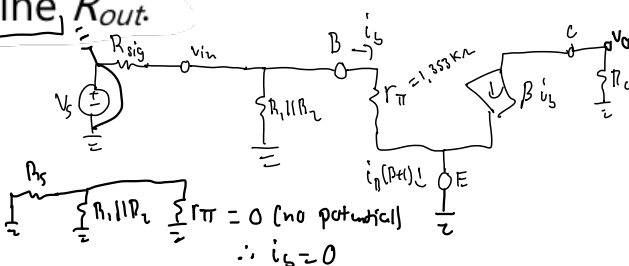
$$\frac{v_o}{v_s} = \frac{-\beta i_b (R_C || R_L)}{i_b r_{\pi} (R_1 || R_2 || r_{\pi}) + R_s} = \frac{-\beta (R_C || R_L) (R_1 || R_2 || r_{\pi})}{r_{\pi} (R_1 || R_2 || r_{\pi}) + R_s}$$

$$\frac{v_o}{v_s} = -36.38$$



$$R_{in} = \frac{v_{in}}{i_{in}} = (R_1 || R_2) || \frac{i_b r_{\pi}}{i_b} = (R_1 || R_2) || r_{\pi} = V_{TH} || \frac{V_T}{I_B} \approx 1.187 \text{ k}\Omega = R_{in}$$

(c) Fine R_{out} .



$$R_o = R_C$$

$$F_{in} = 1 \cdot 10^9$$

(d) Choose some sensible values for the capacitors (think about the role each capacitor plays in this circuit) to estimate the time constants

$$C_1 = 1 \mu F$$

$$C_2 = 10 \mu F$$

$$C_3 = 100 \mu F$$

$$P_2 \approx \frac{1}{C_2(R_0 + R_L)}$$

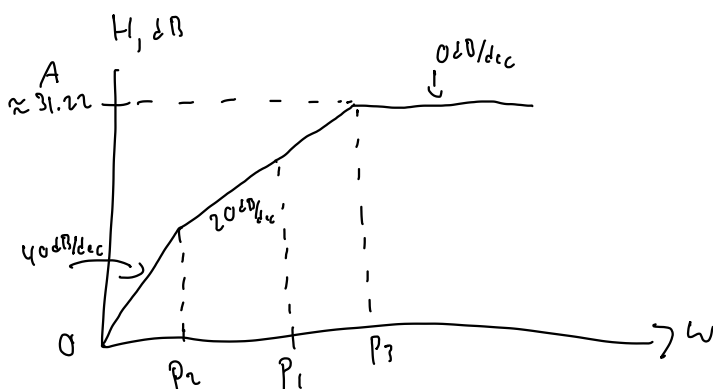
$$P_2 \approx 16.45$$

$$P_1 = \frac{1}{C_1(R_{in} + R_s)}$$

$$P_1 \approx 313.83$$

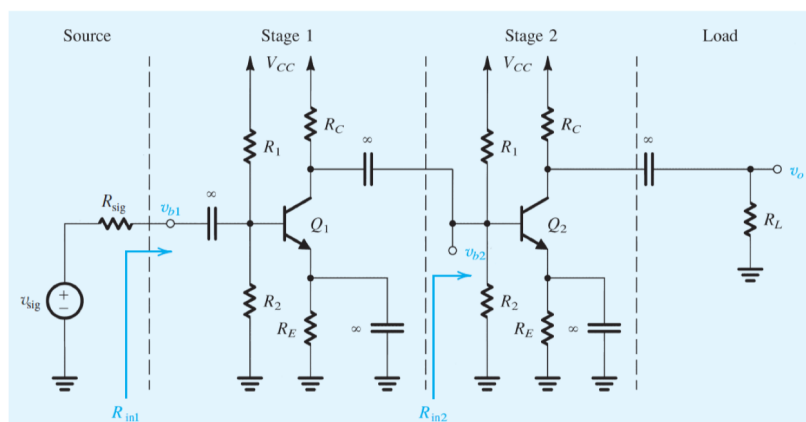
$$P_E: R_E \parallel \left(\frac{V_T}{I_T} \right) = R_E \parallel \left(\frac{r_{\pi}}{\beta + 1} \right)$$

$$P_E = \frac{1}{C_3(R_E \parallel \left(\frac{r_{\pi}}{\beta + 1} \right))} = 780.68$$



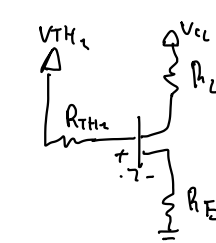
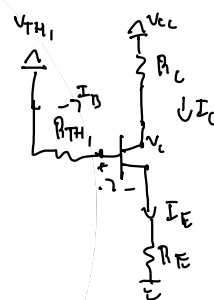
2. The amplifier below consists of two identical common-emitter amplifiers connected in cascade. Observe that the input resistance of the second stage, R_{in2} , constitutes the load resistance of the first stage.

(a) For $V_{CC} = 15$ V, $R_1 = 100$ k Ω , $R_2 = 47$ k Ω , $R_E = 3.9$ k Ω , $R_C = 6.8$ k Ω , and $\beta = 100$, determine the dc collector current and dc collector voltage of each transistor.



Stage 1

Stage 2



$$V_{TH1} = 15 \left(\frac{47}{147} \right) = \frac{215}{44} \text{ V}$$

$$V_{TH2} = \frac{235}{44} \text{ V}$$

$$R_{TH2} = 31.978 \text{ k}\Omega$$

$$R_{TH1} = 31.973 \text{ k}\Omega$$

$$V_{TH1} = I_B R_{TH1} + I_E R_E$$

$$\frac{2007}{440} = I_B (R_{TH1} + (\beta + 1) R_E)$$

$$I_B \approx 9.62 \mu A$$

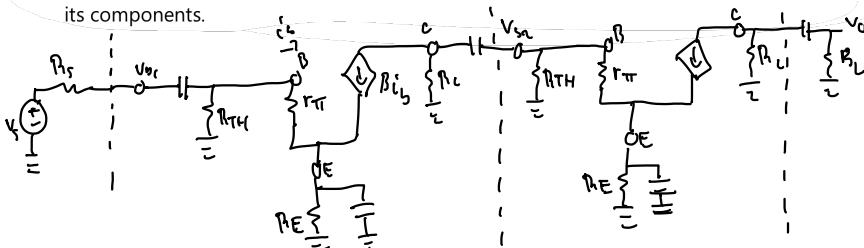
$$I_C = \beta I_B \approx 0.962 \text{ mA}$$

$$V_{CE} - V_{CE} = I_C R_C$$

$$V_{CE} = V_{CC} - I_C R_C = 8.46 \text{ V}$$

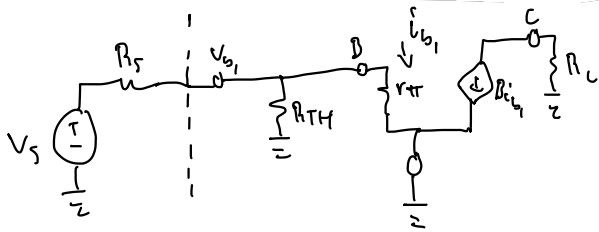
same for stage 2

(b) Draw the small-signal equivalent circuit of the entire amplifier and give the values of all its components.



$$r_{\pi} = \frac{V_T}{I_B} \approx 2.6 \text{ k}\Omega$$

(c) Find R_{in1} and v_{b1}/v_{sig} for $R_{sig}=5\text{ k}\Omega$.



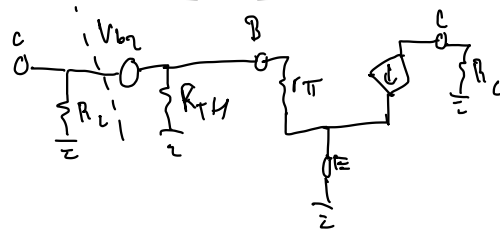
$$v_{b1} = i_b r_{\pi}$$

$$v_s = i_b (R_s + (R_{TH} || r_{\pi}))$$

$$R_{in1} = R_{TH} || r_{\pi} \approx 2.404\text{ k}\Omega$$

$$\frac{v_{b1}}{v_s} = \frac{r_{\pi}}{R_s + (R_{TH} || r_{\pi})} \approx 0.35$$

(d) Find R_{in2} and v_{b2}/v_{b1} .

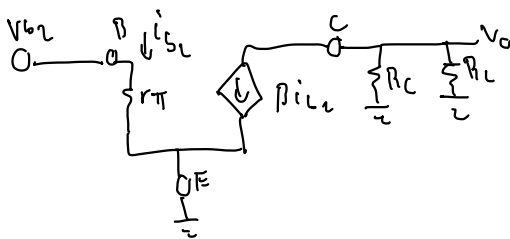


$$R_{in2} = R_{TH} || r_{\pi} \approx 2.404\text{ k}\Omega$$

$$v_{b2} = -\beta i_{b1} (R_L || R_{in2})$$

$$\frac{v_{b2}}{v_{b1}} = \frac{-\beta i_{b1} (R_L || R_{in2})}{i_{b1} r_{\pi}} = \frac{-\beta (R_L || R_{in2})}{r_{\pi}} \approx -68.53$$

(e) For $R_L=2\text{ k}\Omega$, find v_o/v_{b2} .



$$v_o = -\beta i_{b2} (R_L || R_{in2})$$

$$\frac{v_o}{v_{b2}} = \frac{-\beta i_{b2} (R_L || R_{in2})}{i_{b2} r_{\pi}} = \frac{-\beta (R_L || R_{in2})}{r_{\pi}} \approx -54.46$$

(f) Find the overall voltage gain v_o/v_{sig} .

$$\frac{v_o}{v_s} = \frac{v_o}{v_{b2}} \cdot \frac{v_{b2}}{v_{b1}} \cdot \frac{v_{b1}}{v_s}$$

$$\frac{v_o}{v_s} = \left(\frac{-\beta (R_L || R_{in2})}{r_{\pi}} \right) \left(\frac{-\beta (R_L || R_{in1})}{r_{\pi}} \right) \left(\frac{r_{\pi}}{R_s + (R_{TH} || r_{\pi})} \right)$$

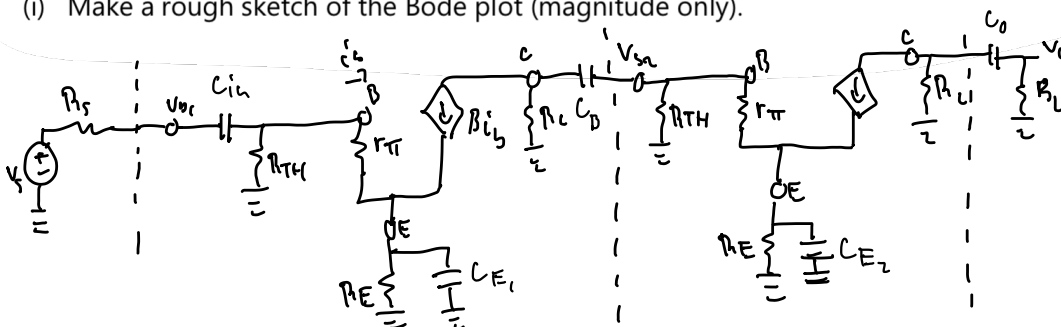
$$\frac{v_o}{v_s} \approx 1426.24$$

(g) Find R_{out} .

$$R_{out} = R_L$$

(h) Choose some sensible values for the capacitors (think about the role each capacitor plays in this circuit) to estimate the time constants

(i) Make a rough sketch of the Bode plot (magnitude only).



$$C_{in} = C_o = C_E1 = 1\text{ }\mu\text{F}$$

$$C_{E1} = C_{E2} = 100\text{ }\mu\text{F}$$

$$P_{in} = \frac{1}{C_{in} (R_s + (R_{TH} || r_{\pi}))}$$

$$\approx 135.1\text{ rad/s}$$

$$P_{E1} = \frac{1}{C_{E1} (R_{E1} || \frac{r_{\pi}}{\beta+1})}$$

$$P_{E2} = \frac{1}{C_{E2} (R_{E2} || \frac{r_{\pi}}{\beta+1})}$$

$$P_{E1} = P_{E2} \approx 391.13\text{ rad/s}$$

double pole

$$P_{CB} = \frac{1}{C_o (R_L || (R_{TH} || r_{\pi}))}$$

$$P_{CB} \approx 543.1\text{ rad/s}$$

$$P_{CO} = \frac{1}{C_o (R_L || R_{in2})}$$

$$P_{CO} \approx 647.1\text{ rad/s}$$

